Hemant Banke

ISI Kolkata, M.Stat. | Goldman Sachs | Quantitative Strategist
(+91)789-879-2172 | hemantbanke5@gmail.com | linkedin.com/in/hemant-banke | github.com/Hemant-Banke

Research Interests: Reinforcement learning, Multi-agent RL, Offline-to-online learning, Long-horizon sequential decision-making

Education

Indian Statistical Institute (ISI Kolkata) June 2023
Master of Statistics (M.Stat.), Specialization in Computational Statistics First division with **Distinction**

- Key Courses: Stochastic Processes, Probability Theory, Pattern Recognition, Statistical Inference, Statistical Computing, Theory of Games and Statistical Decisions, Time Series Analysis

Holkar Science College June 2021
Bachelor of Statistics, Minor in Mathematics and Computer Science First division with **Distinction**

Research Experience

Multi-Agent RL for Defensive Coordination in Soccer | Ongoing, Co-Authored 2026

- Two-team Markov game solved via **offline-to-online minimax self-play**: attacker and defender policies pretrained offline on player-tracking data, then fine-tuned online against an adaptive learned opponent using an OBSO-based reward. Reframes defensive positioning as equilibrium under an adaptive opponent

RL for Goal-Based Wealth Management with Debt Decisions | Ongoing, Co-Authored 2026

- Solving the GBWM long-horizon stochastic optimal control problem via RL over multi-decade decision horizons; extends the standard formulation to heterogeneous asset and debt accounts with voluntary debt actions, expanding the action space beyond prior literature

Concept Drift and Domain Transfer for GPP Estimation | Master's Thesis, ISI Kolkata 2023

- Under Dr. B. Uma Shankar. Studied transfer of supervised models from global ecological data to a low-data target domain (Indian terrain); quantified concept drift (Drifter) and out-of-distribution risk via Direct Loss Estimation

Nowcasting Stock Implied Volatility with Twitter | Extension 2023

- Extended Dierckx et al. (2022), under Dr. Diganta Mukherjee. Day-ahead IV forecasting gains via gradient-boosted ensembles combining Twitter-derived attention/sentiment features with market-data features

Professional Experience

Goldman Sachs | Associate, Quantitative Strategist, AWM Multi-Asset Solutions May 2026 – Present

Wells Fargo | Quantitative Analytics Specialist, Consumer Model Development Center June 2023 – May 2026

- Designed utility (reward) functions for a high-dimensional goal-based RL problem via an **axiomatic approach**: specified desired base properties, derived the class of functions they induce, selected a functional form within it, and analyzed its tradeoffs against planning variables
- Built a **Bradley–Terry–Luce preference model** to rank next-best advice for customers conditioned on their goal plan, the preference-learning model class underlying RLHF reward models
- Solved multi-goal capital allocation as constrained LP, with heterogeneous accounts and debt decisions; sampled within 1800+ dimensional constrained feasibility region via **Hit-and-Run MCMC sampling**
- Built a **high-concurrency Monte-Carlo simulation framework** (Python, OOP, multiprocessing) to backtest allocation strategies over correlated-GBM market paths at scale
- Designed a statistical pipeline for **synthetic data generation**: synthesizing high-complexity customer profiles with non-linear real-world behaviors to improve training density and coverage for downstream financial models
- Estimated heterogeneous treatment effects of digital interactions via **causal inference** (T-Learner with gradient boosting, propensity-score matching)
- Designed and taught 18+ hours of ML curriculum to 80+ analysts; won two Quarterly Recognitions & Spotlight Awards
- Founded initiative MASE-AI, an institutional RAG to develop in-house tutor & work assistant

Projects

Ensemble Methods for Expected Goals (xG) Modeling | ML, Stacked Ensembles, Python

- Under Dr. Deepayan Sarkar. Stacked ensemble (penalized logistic regression, kNN, gradient-boosted meta-learner) for shot-quality estimation from spatial and contextual features (78.5% ROC-AUC); addressed class imbalance via ROSE
- Motivated the multi-agent RL soccer paper above

Skills

- **ML / RL**: Reinforcement Learning (policy gradients, self-play, offline & offline-to-online RL), Preference Learning (Bradley–Terry), Deep Learning, Game Theory, Monte-Carlo & MCMC, Optimization (LP, MILP, Convex), Statistical & Causal Inference, Stochastic Processes
- **Programming**: Python (NumPy, pandas, SciPy, Matplotlib, scikit-learn), R, C++, SQL, Git, Linux

Achievements

- **All India Rank 39** in ISI M.STAT. 2021 exam (top 1%); **All India Rank 84** in IIT-JAM MS 2021 exam (top 0.8%)
- Scored **Top 0.1%** Rank in IIT-JEE Mains 2018 exam (1+ million candidates)
- Full tuition waiver with stipend (Master's) and full scholarship (Bachelor's); graduated both with Distinction
- YC Startup School 2026; Won **$25K** worth AI credits
- Ranked **globally 1st** and **19th** in Manual Trading, IMC Prosperity 4 trials
- Won Inter-school BalVigyaan CS Competition twice; Built skin shader in HLSL with Sub-surface scattering and IBL
- Hackathon Finalist, Code-N-Counter 2020 by Nagarro and Coding Ninjas

Extracurricular Activities

- Built simple Game Engine with C++ & OpenGL at 11; started coding shaders and selling them on Unity3D Asset Store
- Member of Google Developer Group, Facebook Developer Community (2018), E-Cell (2021)
- Computer Programming for Engineers at Wichita State University (2019)
- Started Freelancing web-apps during Bachelor's and got hired as Founding engineer of a startup
- Developed MVP of a two-sided service marketplace with Escrow and Wallet systems supporting cards and crypto
- Doubt Clearing Coach at Cheenta School of Statistics during Master's (2021)